

HDV/HDE Booking Tool

Business Requirement Document — v10

Version: 10.0

Date: 2026-04-22

Based on: BRD v9 (2026-04-22) + Meeting: Open Questions Review 22.04.2026

Prepared by: Telman Nurzhanov (SA)

Changelog v9 → v10

#	Section	Change
1	10. Open Questions	OQ-29 CLOSED: dynamic filter matrix — covered by dynamic characteristics admin tool
2	10. Open Questions	OQ-36 CLOSED: aggregate request status — resolved by dev team
3	10. Open Questions	OQ-NEW-A CLOSED: equipment field list per type — covered by dynamic characteristics admin tool
4	10. Open Questions	OQ-NEW-B CLOSED: equipment/request template — dynamic characteristics pulled per equipment type automatically; no manual template selection
5	10. Open Questions	OQ-NEW-C PARTIALLY CLOSED: SWP role defined — users with SWP role see JDE-sourced requests; users without role use Default WO; JDE trigger status still TBD
6	10. Open Questions	OQ-NEW-E CLOSED: FO delegation via AAD groups (managed outside system); no in-system delegation feature required
7	10. Open Questions	OQ-NEW-F CLOSED: equipment type parameters — covered by dynamic characteristics admin tool

#	Section	Change
8	10. Open Questions	OQ-NEW-N CLOSED: equipment type fields and data examples — covered by dynamic characteristics admin tool
9	10. Open Questions	OQ-NEW-P UPDATED: DataLake JDE jobs run once/day (~05:00 AM); direct JDE API integration proposed as preferred approach; pending JDE team
10	10. Open Questions	OQ-NEW-Q PARTIALLY CLOSED: utilization updated once at end of working day; previous-day data shown in dashboard; satisfies Logistics; Maintenance team frequency TBD
11	10. Open Questions	OQ-NEW-R added: whether long-term rented equipment transferred to another department requires additional CC approval for re-invoicing — to discuss at next BRD review
12	3. Glossary	"Assigned" equipment clarified: bookable by Admin-authorized users with mandatory justification; Admin pre-authorization remains required; FO can decline
13	3. Glossary	Service Work Processor clarified: role grants access to JDE-sourced requests; users without role work via Default WO only
14	4.2	Service Work Processor role definition updated (22.04)
15	4.3	FO item 11: delegation via AAD groups (outside system) — Confirmed/Closed; no in-system delegation feature
16	4.5	Admin item 13: pre-authorization for Assigned equipment — Confirmed; mandatory justification added as additional requirement
17	4.5	Admin item 14: delegation settings — Obsolete; delegation via AAD outside system
18	5.1.1	Equipment Type-Specific Parameters: approach confirmed — dynamic custom attributes per equipment type via Admin Panel tool
19	6.1	FR-NEW-02: Obsolete — FO delegation handled via AAD groups outside system
20	6.2	FR-010 updated: Admin manages custom characteristics per equipment type (dynamic attributes) — Confirmed

#	Section	Change
21	6.2	FR-NEW-04: Confirmed — Admin pre-authorization for Assigned equipment remains valid; mandatory justification added as additional requirement
22	6.2	FR-NEW-68 added: system shows only type-specific dynamic characteristics in request form and search filters
23	6.3	FR-NEW-08 updated: Assigned equipment bookable by Admin-authorized users with mandatory justification; FO can decline
24	6.3	FR-NEW-39: Confirmed — covered by dynamic characteristics admin tool
25	6.12	FR-NEW-62 updated: GIS tooltip to include current-day utilization (Maintenance requirement); refresh interval TBD (see OQ-NEW-Q)
26	8.1	JDE integration: DataLake jobs run once/day; direct JDE API proposed as preferred solution; pending JDE team
27	8.5	DataLake utilization: refresh once at end of working day; previous-day data displayed in dashboard
28	8.5	DataLake GIS: tracker types clarified (Teltonika — real-time; LoRaWAN — 3×/day; GPS-only sensors — 3×/day)

Summary of Changes vs. Original BRD

The table below reflects the most significant scope and design decisions made during requirements gathering sessions (April 2026) relative to the original BRD delivered by the client.

Area	Original BRD	Current Version (v10)
Booking scope	Full booking workflow for both internal (TCO) and external (BP) equipment	Internal and long-term rented equipment only. On-demand BP equipment is displayed as a catalog/showcase — no booking workflow for on-demand BP in Phase 1
External approval chain	CC Owner → External Fleet Owner required	CC Owner role is obsolete. All external booking FRs (FR-051–057) removed from scope

Area	Original BRD	Current Version (v10)
	for BP equipment bookings	
DCT integration	Required: RWA/Change Order creation; tracker data forwarding	Obsolete — external fleet booking removed from scope
PSWS integration	Required: CC Owner identification; user email from White Pages	CC Owner role removed; PSWS used for contact data only
DOA integration	Required: DOA processing of CC Owner bookings	Obsolete — external fleet booking removed from scope
Booking status transitions	Automatic by system datetime (FR-070, FR-071, FR-075)	Manual close only — status changes are triggered by user action, not datetime
FO date editing	Before confirmation only (FR-048 original)	Before and after confirmation; Requestor notified automatically; no re-confirmation required
Equipment classification	Internal fleet / External fleet	Three dimensions: Mobility (self-propelled / unwheeled / stationary) × Ownership (TCO Owned / Long-term rented / BP Showcase) × Usage Status (Assigned / Shared without Conditions / Shared with Conditions)
Equipment card — BP fields	Not described	Unified database for TCO and BP equipment; subset of fields optional for BP
Utilization	Basic monitoring via trackers (FR-099)	Full utilization dashboard: 3 tabs, color gradient, repair-day markers, target values, average, Work Center filter, ownership filter
GIS map	Basic coordinates display (FR-100)	Full GIS iframe integration (MAPH/Atlas team); speed/direction; historical route; tooltip with sensor data and current-day utilization

Area	Original BRD	Current Version (v10)
New roles	—	Transportation Responsible (for unwheeled equipment transport)
New concepts	—	Mobilization period; Default Work Order / Work Center; equipment freeze mechanism; Booking snapshot; dynamic characteristics per equipment type
Fleet structure	One Fleet Owner per fleet	Fleet can have 2+ Fleet Owners with shared team email
Equipment templates	Static admin-managed templates	Dynamic characteristics: Admin configures per-type attributes via Admin Panel tool; shown automatically based on selected equipment type
Assigned equipment	Bookable only by pre-authorized users	Bookable by Admin-authorized users with mandatory justification; FO can decline
FO delegation	In-system delegation with date range	Delegation via AAD groups (outside system); no in-system feature

Status Reference

Status	Description
Confirmed	Requirement confirmed in meetings; no conflicts
Pending	Requirement accepted; details TBD or pending sign-off
Obsolete	Cancelled or replaced by another requirement
Backlog	Confirmed direction; deferred after baseline implementation
Conflict	Contradiction between original BRD and meeting decisions; requires explicit sign-off

1 Introduction

1.1 Purpose

This document defines the business requirements for a booking tool that allows users to create requests for HDV/HDE equipment. Requests may include multiple pieces of equipment, each of which is treated as a separate booking with its own approval workflow.

1.2 Scope

The tool will support bookings for company-owned equipment as well as long-term rented equipment. It will cover request creation, approval flow, status tracking, reporting, and integration with external systems where applicable.

Scope decision (08.04): Booking through the tool is available only for TCO Owned and Long-term rented equipment. On-demand BP (external fleet) is displayed as a catalog/showcase only — no booking workflow for on-demand BP in the system. This is a significant scope reduction vs. the original BRD. See Section 6.7 for detail.

2 Business Context

2.1 Current Challenges

- Lack of centralized booking and collaboration between departments on shared resources
- Low transparency on total number of units and spend
- No standardized process or guidelines
- No Process Owner(s)
- Lack of visibility due to manual utilization monitoring

2.2 Business Objectives

- Standardize the request and booking process.
- Ensure automated routing of approval based on equipment fleet ownership.
- Provide transparency and accountability in the booking process.
- Enable scalability for both internal and partner-provided equipment.
- Process ownership by Logistics.

3 Glossary

Request – a container grouping multiple equipment bookings (one per equipment item).

Regular Request – a request created by Requestor in HDV/HDE booking tool.

Service Work Request – a request created in HDV/HDE automatically via API based on a Work Order created in JDE E1, once the Work Order step reaches a certain status. Visible only to users with the Service Work Processor (SWP) role. *(OQ-NEW-C — PARTIALLY CLOSED 22.04)*

Booking – individual booking within a request. One booking is linked to one unique equipment item. Treated individually within a request.

Equipment – individual asset (HDV/HDE unit) that can be booked through the system. Each equipment item belongs to a fleet and inherits approval responsibility from the corresponding Fleet Owner.

Fleet – logical group of equipment owned by a Fleet Owner (for internal: Maintenance, Construction, TFM, etc.; for external: business partner's name). Each fleet has a unique name assigned at creation.

Fleet Owner (FO) – person responsible for approvals within a fleet. One fleet can have 2+ Fleet Owners (e.g. shift workers). A fleet has a shared team email used for system notifications. Delegation is managed via AAD group membership (outside the system).

Approver – Fleet Owner and/or Cost Center (CC) owner, who confirms/declines bookings.

Mobilization (*Мобилизация*) – the period of preparation/movement of equipment before it arrives at the work site. Booking period starts from mobilization start, not from arrival. Added during requirements gathering.

Equipment categories — Usage Status:

- **Assigned** – equipment assigned to a specific department; visible to all users. Bookable only by **Admin-authorized users** with a mandatory **Justification** field; Fleet Owner can decline. Not available in the general shared pool. *(Закреплённая техника. Clarified 22.04: Admin pre-authorization required; mandatory justification added as additional requirement)*
- **Shared without Conditions** – equipment available for general booking without any preconditions. *(Общедоступная без условий)*
- **Shared with Conditions** – equipment available for booking, but the Requestor must provide a **Justification** field. Used for specialized equipment where FO needs context (e.g. specific

compressors, chemical tanks). Fleet Owner can recall any booking at any time regardless of this status — recall is a separate mechanism. *(Общедоступная с условиями; обоснование обязательно. Решение закрыто 13.04)*

Equipment Ownership Types:

- **TCO Owned** – internally owned by TCO.
- **Long-term rented** – rented long-term; treated as internal fleet for booking purposes. Managed by internal TCO Fleet Owner (not by BP). Booking process is identical to TCO Owned equipment.
- **On-demand rented (BP Showcase)** – rented on demand; displayed as catalog/showcase only in Phase 1.

Dynamic Characteristics *(Динамические характеристики)* – custom equipment attributes configured per equipment type via Admin Panel tool. These characteristics are shown automatically in the equipment card, request form, and search filters based on the selected equipment type. Replaces the static template approach. *(Added 22.04)*

Utilization *(Утилизация)* – daily efficiency of equipment usage; reflects how effectively the company manages its equipment.

- Tracked by **mileage** (wheeled equipment) or **engine hours** (other equipment types), or **both** depending on equipment category.
- Each Fleet Owner independently defines the **100% utilization threshold** for each equipment type (maximum moto-hours or km per 24 hours = 100%).
- **Maximum Utilization Capacity per Shift** — maximum possible utilization within a working shift; serves as the 100% reference point.
- Utilization percentage is calculated per day, month, or any selected period.
- Some equipment types have **both** mileage and engine hours parameters. Planned values are entered by FO; actual values are sourced from trackers only (not manually editable). *(Updated 13.04)*
- Utilization is sourced from **trackers only**, independent of whether a booking exists in the system. This allows detecting equipment usage not linked to any system request. *(Updated 17.04)*
- Utilization dashboard data is **refreshed once at the end of the working day**; dashboard shows data for the **previous day**. Satisfies Logistics requirements. Maintenance team refresh frequency — TBD *(OQ-NEW-Q — PARTIALLY CLOSED 22.04)*.

Utilization Calculation Formulas *(Updated 17.04):*

Metric type	Formula
Engine hours only	$\text{utilization} = \text{engine_time} / \text{target_mh_per_day}$
Mileage only	$\text{utilization} = \text{mileage} / \text{target_km_per_day}$
Both (hours + km)	$\text{utilization} = \max(\text{engine_time} / \text{target_mh_per_day}, \text{mileage} / \text{target_km_per_day})$
Cap rule	If utilization > 1 → display as 1 (100%)

Business goal (14.04): Eliminate equipment downtime. Primary metric — how effectively equipment is used.

Utilization Dashboard tabs *(Added 17.04):*

- **Engine hours** — engine hours utilization only. Applicable to HDE equipment.
- **Mileage** — mileage utilization only. Applicable to HDV equipment.
- **Combined** — maximum of the two metrics (Combined). For equipment with both engine hours and mileage.

Utilization average calculation *(Added 17.04):*

- Average is calculated over the **selected date range**.
- **Repair days are excluded** from the calculation denominator (days when equipment had an open JDE maintenance Work Order).
- Days when equipment stood idle (not on repair) **are included** in the calculation.

Priority *(Updated 17.04):*

Priority is an **attribute of the Work Order (Request)** as a whole, not of an individual Work Center step and not of a specific equipment item. All Work Center steps within the same Work Order share the same priority as the parent WO. (OQ-NEW-I — CLOSED)

Department in reports *(Updated 17.04):*

When generating department-level reports, the department is determined by the **Equipment's** department assignment — not by the Fleet Owner's department. Fleet Owner and Equipment

may belong to different departments. (OQ-NEW-J — CLOSED)

Criticality (*Критичность*) – equipment marked as critical requires **security escort (service de sécurité)** during transportation. Not a maintenance priority indicator. Displayed in equipment card. (Updated 13.04)

Work Center (*Рабочий центр / Шаг*) — scope of work to be performed within a Work Order; called a "step" because work is performed sequentially. Attribute of the Work Order Step, not of the equipment unit. (Added 16.04)

Key constraints:

- One equipment type maps to exactly **one** Work Center.
- One Work Center can be served by **multiple** equipment types.

Preliminary Work Center codes (subject to update by Asylbek):

Code	Description
LRG	Группа ГПМ
MGT	Механики по дизелям
BHOE	Экскаватор, трактор ОТО
DTRK	Самосвал ОТО
FTRK	Грузовик ОТО
GDRV	Грузовик с краном ОТО
HYDR	Водоструйная очистка ОТО
OILT	Oil Tanker
VTRK	Вакуумная машина ОТО
CDECK	Транспортная платформа ОТО
CRANE	Кран ОТО
FORKL	Вилочный погрузчик ОТО
GMLOG	Механик передвижного оборудования
MLIFT	Подъёмник ОТО

Code	Description
MSINS	ППУ

Note: Logistics subdivision does not operate Work Centers in JDE. WO and WC fields are optional for Logistics. (OQ-NEW-G — CLOSED 16.04)

Historical route (*Маршрут движения*) – display of equipment movement on the GIS map for a selected date range; data sourced from trackers via DataLake. Displayed for up to **7 days**. (OQ-NEW-K — CLOSED 20.04)

4 Stakeholders & Roles

4.1 Requestor

All users in TCO network will by default have Requestor role to book equipment of the internal fleet (TCO Owned + Long-term rented HDV/HDE). On-demand external (BP) equipment is displayed as a showcase/catalog only — booking of external equipment is handled in DCT, not in this tool.

Requestors are allowed to:

1. Create a Regular Request — Confirmed
2. Submit the Regular Request — Confirmed
3. Prolong booking — Confirmed
4. Submit feedback on the booked equipment — Confirmed
5. Monitor utilization of the booked equipment for the booking period, if equipment has tracker(s) — Confirmed
6. ~~Terminate booking (only external fleet)~~ — Obsolete (external fleet booking removed from scope)
7. Close booking by pressing "Close" button (manual close only) — Confirmed (*Added*)
8. Book Assigned equipment if authorized by Admin; mandatory justification required; Fleet Owner approves/declines — Confirmed
9. View own request history (completed requests) — Confirmed (*Added 17.04*)

4.2 Service Work Processor

Clarified (22.04): The Service Work Processor role is a **dedicated system role** that grants access to JDE-sourced requests (Service Work Requests). Users with the SWP role can view, manage, and submit Service Work Requests created automatically from JDE Work Orders. Users

without the SWP role do not see JDE-sourced requests — they work with Regular Requests using the Default Work Order option. Formal confirmation from TCO on JDE trigger status still required (OQ-NEW-C — *PARTIALLY CLOSED*).

Service Work Processors are allowed to:

1. View Service Work Requests (JDE-sourced, auto-created as Draft) — Confirmed
2. Submit Service Work Requests to Fleet Owner for processing — Confirmed
3. View SWRs grouped by Work Orders — Pending (OQ-NEW-C)

4.3 Fleet Owner

For internal Fleet Owners, access provision is managed by AAD group. There can be 2+ AAD groups to manage internal Fleet Owners (separate group for each fleet created by Administrator).

One fleet can have **two or more Fleet Owners** (e.g. shift workers). Both owners are displayed in the equipment card. Each fleet has a **shared team email** address used as a single notification target (alongside personal emails via AD group). When a Fleet Owner changes role, Administrator reassigns the fleet without rebuilding the fleet structure. (*Updated 13.04*)

Fleet Owners are allowed to:

1. Create new equipment under own fleet — Confirmed
2. Update equipment (only parameters allowed for editing by Administrator) — Confirmed
3. Delete equipment (soft delete) — Confirmed
4. Confirm / decline requests to book equipment under own fleet — Confirmed
5. Update booking period before or after confirmation; Requestor is notified automatically; no re-confirmation required — Confirmed (*Updated 10.04*)
6. Change equipment in booking — Confirmed
7. Terminate already confirmed bookings (only internal fleet) — Confirmed
8. Freeze / unfreeze own equipment from booking for a period or without end date — Confirmed
9. Monitor utilization of own equipment via utilization dashboard — Confirmed
10. Press "Mobilization started" button to fix mobilization start time — Confirmed (*Added*)
11. Delegate FO privileges via AAD group membership (managed by AAD Admin outside the system; no in-system delegation feature) — Confirmed/Closed (*Updated 22.04 — OQ-NEW-E CLOSED*)
12. One user can be an owner of several fleets — Confirmed (*Added*)
13. One fleet can have multiple Fleet Owners; both displayed with shared team email — Confirmed (*Updated 13.04*)

4.4 CC Owner (incl. DOA)

Obsolete (10.04): CC Owner confirmation is NOT required for Long-term rented equipment. CC Owner is applicable only to on-demand external fleet booking, which is handled in DCT system.

CC owners are allowed to:

1. ~~Confirm / decline external fleet equipment booking request~~ — Obsolete

4.5 Administrator

Administrator role will be provided to process owners – Logistics. Access provision is managed by Azure Active Directory (group). Users with an Administrator role are allowed to:

1. Create/edit internal fleet and link with AAD group — Confirmed
2. Create/edit external fleet and manage external fleet owners — Confirmed
3. Delete fleet — Confirmed
4. Create internal fleet equipment — Confirmed
5. Update equipment's parameters (extended edit) — Confirmed
6. Delete equipment (soft delete) — Confirmed
7. Manage dynamic characteristics per equipment type via Admin Panel tool — Confirmed (*Updated 22.04, replaces static template approach*)
8. Manage 'Request' template — Obsolete (replaced by dynamic characteristics per equipment type) (*Updated 22.04*)
9. Manage 'Booking' template — Pending
10. Access to custom reportings — Confirmed
11. Access to utilization dashboard(s), if equipment has tracker(s) — Confirmed
12. Manage all configurable system parameters via Admin Panel — Confirmed (*Added*)
13. Authorize specific users to book Assigned equipment — Confirmed (*Updated 22.04 — remains valid; mandatory justification added as additional requirement*)
14. ~~Manage delegation settings~~ — Obsolete (*Updated 22.04 — delegation via AAD groups outside system*)
15. Manage fleet shared team email — Confirmed (*Updated 13.04*)
16. Manage reference/handbook values (equipment classes, categories, subcategories, manufacturers, models, companies, locations, work centers, divisions, groups, departments, units) via Admin Panel — Confirmed (*Added 16.04*)

4.6 Transportation Responsible (*Мобилизация/Транспортировка*)

A dedicated system role for the team responsible for transporting non-motorized (Unwheeled) equipment (Heavy Ops / Maintenance team). Added during requirements gathering.

This role is allowed to:

1. Receive notification about transportation request for Unwheeled equipment — Confirmed
2. Confirm or decline the transportation request — Confirmed

5 Entities

5.1 Equipment

All equipment must have a mandatory attribute — fleet. The attribute determines the equipment template, who can create/edit/delete it, and the approval and processing flow.

Fleet naming (10.04): Each fleet must have a unique **name** assigned at creation. Equipment assignment is linked to a specific employee; when a Fleet Owner changes, the Administrator manually re-assigns the fleet.

Fleet Owners display (13.04): Equipment card displays **up to 2+ Fleet Owners** alongside their **shared team email**.

Work Center assignment (16.04): Each equipment type is assigned to **exactly one Work Center**. Managed via Admin Panel reference.

Reference/Handbook policy (16.04): The following are handbook-managed via Admin Panel:

- Equipment classes (HDE, HDV)
- Categories and subcategories
- Manufacturer, Model
- Companies (TCO White Pages)
- Locations
- Work Centers
- Divisions, Groups, Departments, Units

Dynamic characteristics per equipment type (22.04): Admin configures custom attributes (dynamic characteristics) per equipment type via Admin Panel tool. These attributes are displayed automatically in the equipment card, request creation form, and search filters based on the selected equipment type. Base data — from the provided equipment database (DB from TCO). This approach replaces the static template concept.

Unified database (confirmed 20.04): TCO-owned and BP (long-term rented) equipment share a single database. For BP equipment, a subset of fields is optional (null allowed):

Field	TCO Owned	BP (Long-term rented)
Category / Subcategory	Mandatory	Mandatory
Manufacturer / Model	Mandatory	Mandatory
Ownership	Mandatory	Mandatory
TCO (TШO) number	Mandatory	Optional
License plate	Optional	Optional
VIN / Serial number	Optional	Optional
Year of manufacture	Recommended	Optional
Service zone	Optional	Optional
Repair company	Recommended	Recommended

Note on repair company (20.04): It is recommended to store the repair company for all equipment. A single unit may have different contractors for different work types (e.g. hydraulics — Energstim; extraction — DAS).

Operational Status removed: «Операционный статус» (Operational Status) is **removed** from equipment card parameters. Remaining status fields: repair status and freeze status. (*Updated 16.04*)

Internal fleet equipment parameters:

1. Basic fixed info (TCO ID, Description, Service Zone, Class/Category, License Plate, Serial number, etc.) — Confirmed / Pending details
2. Dynamic parameters: Equipment Usage Status, **Cost Center**, CC/Division/Group/Department/Unit, Assigned location, Maintenance BP, Contact number / email, etc. — Confirmed

3. **Ownership field** — «Принадлежность»: **TCO Owned / Long term rented** — Confirmed (*Added 16.04*)
4. **Freeze fields** (*Added 16.04*):
- **Заморозка** (freeze flag)
 - **Причина заморозки** (freeze reason)
 - **Дата завершения заморозки** (freeze end date; empty = indefinite)
- Visible and editable by FO and Admin. (FR-NEW-49)
5. Equipment specifications (picture, Chassis type, Drive type, Loading capacity, etc.) — Confirmed
6. Trackers' IDs (WIALON, IVMS, Vega, etc.) — Confirmed. Multiple trackers per unit supported (*Added*)
7. Planned engine hours (per shift/day) — editable by FO — Confirmed (*Added*)
8. Planned daily mileage (km/day) — editable by FO — Confirmed (*Added*)
9. Actual engine hours / mileage — sourced from tracker only; not manually editable — Confirmed (*Added*)
10. **Dynamic characteristics** — custom attributes per equipment type, configured via Admin Panel tool (*Added 22.04*)

Equipment card UI (13.04–17.04):

- **Booking calendar**: visual calendar showing booked periods
- **Criticality flag**: displayed in card
- **Sharing condition**: Assigned / Shared without Conditions / Shared with Conditions — displayed in card (*Figma update pending*)
- **Work schedule display**: split into day shift / night shift blocks
- **Weekend coverage**: default shift 06:00–06:00; Long Term — 07:00–07:00
- **Multiple photos**: supports uploading multiple photos per unit
- **Sidebar**: Comments block in sidebar
- **Tracker block**: add new trackers from equipment card UI
- **Schedule slider**: slider control for selecting time ranges in work schedule block (*Added 16.04*)

GIS map integration in equipment list (17.04):

- Equipment list (FO view) includes **checkboxes** to select units for display on the GIS map
- Map is rendered by GIS team (MAPH/Atlas) as an **iframe**; system passes equipment identifiers as parameters
- See FR-NEW-61–FR-NEW-67

License plate and TCO number rules (13.04):

- **License plate:** present on most wheeled equipment; may be absent on stationary units.
- **TCO number:** assigned at purchase; long-term rented equipment may not have one.

Equipment Classification — three dimensions:

Dimension	Values
Mobility	Self-propelled / Not self-propelled (Unwheeled) / Stationary
Ownership	TCO Owned / Long-term rented / On-demand rented (BP Showcase)
Usage Status	Assigned / Shared without Conditions / Shared with Conditions

Note (08.04): Mobility category NOT displayed in Requestor UI.

Note (08.04): Stationary HDE excluded from Requestor search.

Note (13.04): Shared with Conditions differs only in presence of mandatory Justification field.

Note (22.04): Assigned equipment differs from Shared with Conditions in that it is not available in the general shared pool; bookable only by Admin-authorized users; requires mandatory justification; FO more likely to decline.

Equipment availability display rules:

- Equipment under maintenance shown grayed out; label "Under maintenance until [Planned date from JDE]"
- Repair status sourced from JDE → DataLake integration — Pending details

5.1.1 Equipment Type-Specific Parameters (*Updated 22.04*)

Approach confirmed (22.04): Equipment type-specific parameters (dynamic characteristics) are configured via the Admin Panel tool. Admin can add custom attributes for each equipment type. These attributes appear automatically in the equipment card, request form, and search filters. Base data sourced from TCO equipment database provided by the client. This approach replaces the earlier concept of static, manually curated field lists per type.

Scope of work: Development of the Admin Panel tool for managing dynamic characteristics. Filters in the request/search form display only characteristics relevant to the selected equipment type.

5.2 Booking

Booking is linked to a single equipment item and has a booking range (datetime). Several bookings form a single request.

Architecture decision (09.04): One Request → multiple RequestItems (one per equipment); each independently processed by FO.

5.2.1 Internal Booking

Booking of TCO Owned (including Long-term rented) equipment. Requires only FO confirmation.

Mobilization (09.04): Booking period starts from mobilization start. FO presses "Mobilization started" to fix actual start.

Updated (10.04): FO can adjust booking period before and after confirmation. Requestor notified automatically.

5.2.2 External Booking

Scope change vs. original BRD: On-demand BP equipment is NOT bookable in Phase 1. Displayed as catalog/showcase only.

- On-demand rented (BP Showcase) — catalog view only.
- Long-term rented — treated as internal fleet.
- ~~Full external booking workflow (CC Owner → External FO)~~ — Obsolete

5.2.3 Booking Snapshot

At booking confirmation, system saves a snapshot of key equipment attributes. — Pending (OQ-37)

5.3 Request

5.3.1 Regular Request

Requestor can add **1 or more** equipment items from any fleet; each is treated as a separate booking.

Request fields (08.04):

- **Work Order (JDE)** — mandatory for Maintenance / Railroad / Operations. One WO per request.
- **Work Center (WC)** — mandatory for JDE-based fleets; **optional for Logistics**.
- **Location** (text input) — for SCM Logistics replaces WO.
- **Work Description** — mandatory, ~50 chars.

- **Comments** — optional.
- **Priority** (P1–P4) — attribute of the Request / Work Order. (*OQ-NEW-I — CLOSED*)

WO pre-fill flow (16.04): Block "Выбрать Work Order из JDE" on Submit Request page. See FR-NEW-48.

Default WO / WC (17.04): For teams not using JDE (Logistics and others) — "Default Work Order" checkbox available; no JDE number required. Analogously "Default Work Center" for teams without WC. See FR-NEW-51.

Request constraints (03.04):

- Booking horizon: 1 month ahead (configurable).
- Max booking duration: 1 week (configurable).

5.3.2 Service Work Request (within Work Order in JDE E1)

Partially clarified (22.04): Auto-created as Draft when WO step reaches a defined status in JDE. Visible only to users with the **Service Work Processor** role. SWP reviews and submits to FO. Users without the SWP role do not see SWRs and use the Default Work Order option instead. Formal confirmation of JDE trigger status required (*OQ-NEW-C — PARTIALLY CLOSED*).

Required WO data fields from DataLake (Updated 16.04):

Field	Description / Notes
Work Order number	Unique identifier in JDE E1
Work Order name	Examples needed from TCO
Status	Clarify list of valid statuses
Status description	Text description (<i>Added 16.04</i>)
Priority	Priority value in JDE E1 — attribute of the WO
Work Center (Step code)	Type of work; e.g. "CRANE", "DTRK"
Step name	Text name of the step
Step volumes	Required quantity of equipment units per step (<i>Added 16.04</i>)
Date range	Start date / End date of the WO step (<i>Added 16.04</i>)

6 Functional Requirements

6.1 User & Role Management

ID	Requirement	Status
FR-001	System supports role-based access control	Confirmed
FR-002	All TCO network users have Requestor access (with RWA initiator restriction for BP booking)	Obsolete — superseded by updated Requestor definition
FR-003	Administrators, Service Work Processors defined by AAD groups	Pending (SWP trigger status — OQ-NEW-C)
FR-004	CC Owner / DOA privileges via PSWS/DOA integration	Obsolete
FR-NEW-01	All time-based parameters configurable via Admin Panel (not hardcoded)	Confirmed
FR-NEW-02	FO can delegate privileges to any TCO employee for a date range; auto-revoke upon expiry	Obsolete (22.04 — <i>delegation via AAD groups outside system; no in-system feature required</i>)

6.2 Admin Panel

ID	Requirement	Status
FR-005	Admin creates fleets with unique names	Confirmed
FR-006	Admin links internal fleet with AAD group	Confirmed
FR-007	Admin manages external fleet owners	Confirmed
FR-008	Admin updates existing fleets	Confirmed

ID	Requirement	Status
FR-009	Admin deletes fleets	Confirmed
FR-010	Admin configures dynamic custom characteristics per equipment type via Admin Panel tool; these characteristics appear automatically in equipment card, request form, and search filters based on equipment type <i>(Updated 22.04 — replaces static template concept)</i>	Confirmed
FR-011	Admin manages request template	Obsolete (22.04 — replaced by dynamic characteristics per equipment type)
FR-012	Admin manages booking template	Pending
FR-013	Admin creates equipment (internal fleet)	Confirmed
FR-014	Admin edits equipment (extended edit)	Confirmed
FR-015	Admin deletes equipment	Confirmed
FR-016	Admin accesses all dashboards and reports	Confirmed
FR-NEW-03	Admin configures: FO timeout (24h/48h), booking horizon, max duration	Confirmed
FR-NEW-04	Admin authorizes specific users to book Assigned equipment	Confirmed (22.04 — remains valid; mandatory justification added as additional requirement)
FR-NEW-42	Admin sets and updates fleet shared team email	Confirmed

ID	Requirement	Status
FR-NEW-50	Admin manages all reference/handbook values via Admin Panel	Confirmed
FR-NEW-68	System dynamically shows only type-specific characteristics (dynamic attributes) in request creation form and equipment search filters, based on the equipment type selected by the Requestor. <i>(Added 22.04)</i>	Confirmed

6.3 Equipment Management

ID	Requirement	Status
FR-017	Both internal and external FO can create equipment under own fleet	Confirmed
FR-018	FO must indicate fleet of equipment; only own fleets selectable	Confirmed
FR-019	FO can edit allowed equipment parameters; one user can own several fleets	Confirmed
FR-020	FO can delete equipment (soft delete)	Confirmed
FR-021	FO can freeze/unfreeze equipment for a period or indefinitely	Confirmed
FR-022	Requestor / SWP can submit feedback on equipment with confirmed booking	Confirmed
FR-NEW-05	FO uploads multiple photos; Requestor sees them in request form	Confirmed
FR-NEW-06	Repair status from JDE → DataLake displayed in search and equipment list	Pending
FR-NEW-07	Stationary HDE excluded from Requestor search	Confirmed
FR-NEW-08	Assigned equipment: visible to all users; bookable only by Admin-authorized users with mandatory justification; FO can approve or decline.	Confirmed

ID	Requirement	Status
FR-NEW-09	Shared with Conditions has visual color marker	Confirmed
FR-NEW-40	Equipment card (FO view) has booking calendar visual	Confirmed
FR-NEW-41	Multiple trackers per unit; FO can add trackers from equipment card UI	Confirmed
FR-NEW-49	Freeze fields on equipment card: Заморозка, Причина заморозки, Дата завершения заморозки	Confirmed

6.4 Request Management

ID	Requirement	Status
FR-023	Requestor can create a Request	Confirmed
FR-024	Request has unique ID and metadata	Confirmed
FR-025	Requestor can view request details and status	Confirmed
FR-026	Request statuses: Draft, Submitted, In Progress, Completed	Confirmed (OQ-36 — <i>CLOSED 22.04; resolved by dev team</i>)
FR-027	Requestor can edit/cancel draft before submission	Confirmed
FR-028	Service Work Requests auto-created with 'Draft' status; visible to SWP role only	Pending (OQ-NEW-C — trigger status TBD)
FR-029	Only SWP can manage (submit) Service Work Requests	Confirmed (<i>Updated 22.04</i>)

ID	Requirement	Status
FR-030	System supports draft saving	Confirmed
FR-NEW-10	On-demand booking only; weekly schedule management out of scope	Confirmed
FR-NEW-11	Requestor selects priority P1–P4 at request creation (priority = WO-level attribute)	Confirmed
FR-NEW-12	WO mandatory for Maintenance/Railroad/Operations; optional for Logistics; Location replaces WO for SCM Logistics	Confirmed
FR-NEW-13	"Work Description" mandatory (~50 chars)	Confirmed
FR-NEW-14	"Comments" optional	Confirmed
FR-NEW-15	Single form → auto-split into individual bookings per equipment item	Confirmed
FR-NEW-16	Partial confirmation: confirmed items proceed independently from declined	Confirmed
FR-NEW-17	Aggregated Request status auto-calculated from RequestItem statuses	Confirmed (OQ-36 — <i>CLOSED</i> 22.04)
FR-NEW-48	"Выбрать WO из JDE" block on Submit Request page; WO list from DataLake; auto-generate draft	Confirmed
FR-NEW-	Default Work Order option for teams not operating in JDE (Logistics and others): checkbox on request form; when selected	Confirmed

ID	Requirement	Status
51	— system does not require a JDE WO number. Analogously Default Work Center for teams without WC. <i>(Added 17.04)</i>	
FR-NEW-64	In the FO booking approval window: display an equipment loading summary by dates — a list of nearest active bookings for the given equipment unit (analogous to a ticket availability view). Allows FO to assess scheduling conflicts before approving. <i>(Added 20.04)</i>	Confirmed

6.5 Booking Management

ID	Requirement	Status
FR-031	Requestor can add/remove equipment items to a request; availability updated	Confirmed
FR-032	System prioritizes internal fleet (business rule)	Confirmed
FR-033	Requestor can add Shared equipment to request	Confirmed
FR-034	System suggests external fleet based on decision matrix	Obsolete
FR-035	Requestor adds external fleet equipment if RWA initiator	Obsolete
FR-036	Justification required when adding external fleet equipment	Obsolete
FR-037	Cost Center required for external fleet equipment	Obsolete
FR-038	Each equipment item in request = separate booking	Confirmed
FR-039	Each booking has unique ID, start/end datetimes	Confirmed
FR-040	System validates availability before booking	Confirmed
FR-041	Equipment attributes displayed in booking	Confirmed
FR-042	Booking lifecycle: Draft, Submitted, Confirmed, Declined, Revoked, Terminated, Completed	Confirmed
FR-NEW-18	Booking start = start of mobilization	Confirmed

ID	Requirement	Status
FR-NEW-19	No mandatory buffer between bookings	Confirmed
FR-NEW-20	Unwheeled equipment: notification displayed to Requestor	Preliminary

6.6 Approval Flow (Internal Fleet)

ID	Requirement	Status
FR-043	Approver assigned automatically based on fleet ownership	Confirmed
FR-044	Internal fleet booking confirmed/declined by FO	Confirmed
FR-045	FO can confirm incoming bookings	Confirmed
FR-046	FO can replace equipment before/after confirmation if booking not yet started	Confirmed
FR-047	FO cannot replace if booking revoked, terminated, or past end date	Confirmed
FR-048	FO can adjust booking date range before or after confirmation; Requestor notified automatically; no re-confirmation required (<i>Updated 10.04 — original: before confirmation only</i>)	Confirmed
FR-049	FO can decline booking; availability updated	Confirmed
FR-050	FO provides reason/comment when declining	Confirmed
FR-NEW-21	FO must respond within timeout. Default: 24h reminder; 48h escalation	Confirmed
FR-NEW-22	If FO has free equipment and declines — mandatory reason required	Confirmed
FR-NEW-23	Reminder sent only to FO of the specific selected equipment	Confirmed

ID	Requirement	Status
FR-NEW-24	FO presses "Mobilization started" to record actual start time	Confirmed

6.7 Approval Flow (External Fleet)

Major scope change (08.04): On-demand BP not bookable. All FR-051..057 are Obsolete vs. original BRD.

ID	Requirement	Status
FR-051..057	External fleet approval chain (CG Owner, External FO, PSWS, DOA)	Obsolete
FR-NEW-25	BP showcase: Requestor can view on-demand BP equipment; no booking action in Phase 1	Confirmed
FR-NEW-26	BP equipment prices/rates NOT displayed	Confirmed
FR-NEW-27	BP populates own catalog cards	Confirmed
FR-NEW-28	"Go to BP catalog" button after FO timeout	Confirmed

6.8 Booking Lifecycle Management

ID	Requirement	Status
FR-058	Requestor/SWP can revoke booking if not yet processed by FO	Confirmed
FR-059	FO can terminate confirmed booking; reason required; availability updated	Confirmed
FR-060	Requestor/SWP can terminate confirmed booking (internal only)	Confirmed
FR-061	Requestor/SWP can extend confirmed booking	Confirmed
FR-NEW-29	Three-step Unwheeled flow: Requestor → FO → Transportation Responsible → FO final	Confirmed

ID	Requirement	Status
FR-NEW-30	If transport unavailable for Unwheeled — FO proposes nearest available date	Pending
FR-NEW-31	Booking can be closed early via "Close early" button	Pending

6.9 Booking/Request Status Management

Conflict RESOLVED (14.04): Auto status transition by datetime cancelled. All status changes are manual only.

ID	Requirement	Status
FR-062	Booking → 'Submitted' once Request submitted	Confirmed
FR-063	Booking → 'Confirmed' once confirmed by FO	Confirmed
FR-064	Booking → 'Declined' once declined by FO	Confirmed
FR-065	External booking → 'Partially Confirmed' by CG Owner	Obsolete
FR-066	External booking → 'Declined' by CG Owner	Obsolete
FR-067	Booking → 'Submitted' once extended by Requestor/SWP	Confirmed
FR-068	Booking → 'Revoked' once revoked	Confirmed
FR-069	Booking → 'Terminated' once terminated	Confirmed
FR-070	Auto-transition to 'In Progress' by datetime	Obsolete — manual close (see FR-NEW-32)
FR-071	Auto-transition to 'Completed' by datetime	Obsolete — manual close (see FR-NEW-32)
FR-072	Request → 'Draft' when saved as draft	Confirmed
FR-073	SWR → 'Draft' when imported from JDE	Pending (OQ-NEW-C)
FR-074	Request → 'In Progress' when first booking (excl. Declined) goes In Progress	Confirmed

ID	Requirement	Status
FR-075	Request → 'Completed' by datetime	Obsolete — see FR-NEW-44
FR-NEW-32	Booking closure is manual only — "Close" button by Requestor or FO	Confirmed
FR-NEW-33	At close: Requestor inputs actual start/end time for utilization analytics	Confirmed
FR-NEW-44	Request → 'Completed' when last active (non-Declined) booking closed	Confirmed

6.10 Notifications

ID	Requirement	Status
FR-076	System notifies SWP when SWR auto-created	Pending (OQ-NEW-C)
FR-077	System notifies FO when request submitted	Confirmed
FR-078	System notifies FO when booking revoked	Confirmed
FR-079	System notifies Requestor/SWP of FO decision	Confirmed
FR-079a	System notifies Requestor when FO updates booking period	Confirmed
FR-080	System notifies Requestor/SWP when FO terminates booking	Confirmed
FR-081	System notifies FO when Requestor extends booking	Confirmed
FR-082..090	External fleet notification chain	Obsolete
FR-NEW-34	24h reminder to FO; 48h escalation to manager	Confirmed
FR-NEW-35	Notification to Transportation Responsible for Unwheeled booking	Confirmed
FR-NEW-36	FO notified when Transportation Responsible decides	Confirmed

ID	Requirement	Status
FR-NEW-43	Notifications delivered to fleet shared team email (and/or AD group personal emails)	Confirmed

6.11 Search & Reporting

ID	Requirement	Status
FR-091	Requestor/SWP can search and filter own requests	Confirmed
FR-092	Approver (FO) can view all pending and completed approvals	Confirmed
FR-093	Admin can view all requests	Confirmed
FR-094	System generates reports on Requests/Bookings/Equipment with filters	Confirmed
FR-095	Admin can view/download all reports	Confirmed
FR-096	FO can view/download own approval reports	Confirmed
FR-097	FO can view/download own equipment reports	Confirmed
FR-098	SWP (and FOs) can view SWR grouped by Work Orders	Pending (OQ-NEW-C)
FR-NEW-37	Audit trail and history on demand	Confirmed
FR-NEW-38	Search: TCO equipment number + model mandatory; госномер if present	Confirmed

ID	Requirement	Status
FR-NEW-39	Dynamic search filters by equipment type — shown automatically based on dynamic characteristics configured per type via Admin Panel (Confirmed 22.04 — OQ-29 CLOSED)	Confirmed
FR-NEW-45	Dedicated "Completed Requests" page for Requestor, SWP, FO, Admin	Confirmed
FR-NEW-46	Utilization reports: by day / department / equipment unit	Pending details
FR-NEW-47	Work center reports (group by WC code)	Pending details
FR-NEW-52	Requestor has access to «История заявок» (Request History) — a dedicated page listing all completed requests by this Requestor. Minimum columns: request number, equipment type, booking period, status, Work Order number. (Added 17.04)	Confirmed

6.12 Dashboards / Map

ID	Requirement	Status
FR-099	System displays utilization and coordinates from DataLake/PI (tracker data)	Confirmed
FR-100	System displays digitized location of Work Orders from DataLake (JDE E1)	Confirmed
FR-NEW-53	Utilization dashboard has three tabs : «Моточасы» (engine hours) / «Километраж» (mileage) / «Комбайн» (max of both). Active tabs depend on equipment type: HDE → Моточасы only; HDV → Километраж only; mixed equipment → all three. (Added 17.04)	Confirmed
FR-NEW-54	Utilization cells use a color gradient : 0% — red → 50% — yellow → 100% — green. Smooth gradient, not discrete steps. (Added 17.04)	Confirmed

ID	Requirement	Status
FR-NEW-55	Repair days are marked in the dashboard with a wrench icon (not a color) to distinguish from low-utilization days. Repair data sourced from JDE (see Section 8.1): <code>commitment date</code> = start of repair, <code>planned date</code> = expected end, <code>completed date</code> = actual end. Repair days are excluded from average utilization calculation . Manual repair input for non-JDE equipment. <i>(Added 17.04)</i>	Confirmed (manual input)
FR-NEW-56	Utilization dashboard includes a Target value column per equipment: maximum daily utilization threshold set by FO (e.g. 2 000 engine hours / 1 500 km per day). Displayed first after equipment name/type columns. <i>(Added 17.04)</i>	Confirmed
FR-NEW-57	Utilization dashboard includes an Average utilization column for the selected period (repair days excluded). Table is sortable by this column . <i>(Added 17.04)</i>	Confirmed
FR-NEW-58	Utilization dashboard supports two time granularity views : «По дням» (daily, current default) / «По месяцам» (monthly average per cell). User toggles via a control above the table. <i>(Added 17.04)</i>	Confirmed
FR-NEW-59	Utilization dashboard left sidebar includes an Ownership filter : TCO Owned / Long-term rented. <i>(Added 17.04)</i>	Confirmed
FR-NEW-60	Utilization data is sourced from tracker telemetry only , independent of bookings. Allows detecting equipment usage not linked to any booking. <i>(Added 17.04)</i>	Confirmed
FR-NEW-61	In the FO Equipment section, equipment list includes checkboxes to select units for display on the GIS map. The GIS map is rendered as an iframe by the GIS team (MAPH/Atlas); system passes selected equipment identifiers as URL parameters. Map supports expand button for full-screen view. Search in equipment list by ТШО-номер. <i>(Added 17.04)</i>	Confirmed (GIS team involvement required)

ID	Requirement	Status
FR-NEW-62	GIS map marker tooltip on click: displays all available sensor data received from the last tracker synchronization, plus current-day utilization (Maintenance team requirement). Data fields: speed, heading, coordinates, engine hours, odometer, ignition, battery, current-day utilization value. Utilization refresh interval for map tooltip: TBD (target ~30–60 min; see OQ-NEW-Q). <i>(Updated 22.04 — original 20.04: all sensor data from last sync only)</i>	Pending alignment with GIS team + refresh interval confirmation
FR-NEW-63	Utilization dashboard displays booking count for the selected period alongside the average utilization value. Allows analysis of correlation between equipment load (bookings) and actual utilization. <i>(Added 20.04)</i>	Confirmed
FR-NEW-64	In the FO booking approval window: display an equipment loading summary by dates — a list of nearest active bookings for the given equipment unit. <i>(Added 20.04)</i>	Confirmed
FR-NEW-65	GIS map displays speed and heading (direction) for equipment equipped with GSM trackers. For equipment with LoRaWAN trackers (battery-powered, sync up to 3 times/day): heading is not displayed due to infrequent synchronization. <i>(Added 20.04)</i>	Confirmed
FR-NEW-66	GIS map does not display fuel level . Fuel telemetry data will not be available from the trackers. <i>(Added 20.04)</i>	Confirmed
FR-NEW-67	GIS map supports display of historical route for selected equipment for up to 7 days . Data sourced from tracker telemetry via DataLake. <i>(Added 20.04, OQ-NEW-K CLOSED)</i>	Confirmed (to be aligned with GIS team)

6.13 Audit & Compliance

ID	Requirement	Status
FR-101	System logs all request creation, modifications, approvals, rejections	Confirmed
FR-102	Audit trail accessible for compliance purposes	Confirmed
FR-103	Exportable logs for legal/regulatory audits	Confirmed

7 Non-Functional Requirements

Area	Requirement	Status
Availability	99.5% uptime	Confirmed
Performance	High volume of requests per hour without degradation	Confirmed
Security	RBAC; Azure AD SSO	Confirmed
Compliance	Full audit history of all approvals / rejections / revokes / terminations	Confirmed
Scalability	Support internal and external partner use cases	Confirmed
Usability	Intuitive UI; UX reference: Booking.com	Confirmed
Users	~200 users (per ACR assessment 07.04); No personal data (SPD) stored	Confirmed
Security tier	ACR assessment result: Low Risk (07.04)	Confirmed
Authentication	Azure SSO for internal users; separate account for external BP	Confirmed

8 Integrations

8.1 JDE E1

Work Orders created in JDE E1 are imported into HDV/HDE Booking Tool as Service Work Requests (SWRs) once a defined status is reached. Each Work Order step is imported as a separate SWR. Only steps with approved equipment types (e.g. BHOE, CDECK, DTRK, FORKL, FTRK, GDRV, GMLOG, MLIFT, VTRK, etc.) are included.

Integration approach (Updated 22.04):

Issue identified (22.04): DataLake jobs for JDE run **once per day** (~05:00 AM). A Work Order step approved during the day would not appear in the booking tool until the following morning — unacceptable for operational use.

Preferred solution (22.04): Direct API integration with JDE E1 (bypassing DataLake for WO data). Allows near-real-time synchronization (target: ~5-minute interval).

Action: To confirm with the JDE team whether a 5-minute API interval is technically feasible. (OQ-NEW-P — UPDATED)

Fallback: If direct JDE API is not feasible, evaluate alternative approaches (e.g. more frequent DataLake jobs triggered by JDE events).

Integration path options:

- **Preferred:** JDE E1 → Direct API → HDV/HDE Booking Tool (target: ~5 min interval)
- **Fallback:** JDE E1 → Data Lake → HDV/HDE Booking Tool (current: once/day ~05:00)

JDE WO flow (Updated 13.04): Steps: status ~50 (scheduler review) → status **55** (released to operations). Steps at status 55 → auto-import as SWR Drafts. SWP reviews and submits.

Required Work Order fields from JDE (Updated 16.04):

Field	Description / Notes
WO Number	Unique identifier in JDE E1
WO Name	Examples needed from TCO
WO Status	Exact list of statuses TBD
WO Status Description	Text description (<i>Added 16.04</i>)
WO Priority	Priority = WO-level attribute (<i>OQ-NEW-I — CLOSED 17.04</i>)
Work Center (Step code)	Defines required equipment type
Step Name	Text name of the step
Step Volumes	Required quantity per step (<i>Added 16.04</i>)
Date Range	Start / End date of the WO step (<i>Added 16.04</i>)

Repair/Maintenance data fields from JDE (via DataLake) (*Added 17.04*):

Field	Description
Commitment date	Date when the maintenance WO was committed / accepted
Planned date	Expected completion date of maintenance work

Field	Description
Completed date	Actual completion date (set when maintenance WO is closed)

These fields are used to identify **repair days** in the utilization dashboard (FR-NEW-55). Repair days are excluded from average utilization calculation.

Status: Pending — JDE trigger status TBD; preferred integration path (direct API) pending JDE team confirmation.

Repair status (08.04): Maintenance in progress / planned completion date displayed in Requestor's equipment search from DataLake. — Pending details.

8.2 DCT (Digital Contractor Timesheet)

Status: Obsolete — external fleet booking removed from scope.

- ~~RWA / Change Order Creation~~ — Obsolete
- ~~Tracker's data receipt~~ — Obsolete

8.3 PSWS (TCO White Page)

Status: Confirmed — White Pages used for auto-populating user contact data. CC Owner role not applicable.

8.4 DOA (Delegation of Authority)

Status: Obsolete — CC Owner role not applicable.

8.5 DataLake

Data Source and Utilization (22.04)

Utilization data is obtained from **Cosmos DB via API**, sourced from **IVMS** and **WIALON** trackers. The equipment **Tracker ID** is used to correlate and merge utilization and location records within the Data Lake.

Utilization refresh frequency (22.04):

- Utilization data is updated **once at the end of the working day**.
- Dashboard shows **previous day's** data.
- This frequency satisfies Logistics team requirements.

- Maintenance team frequency requirements — TBD (OQ-NEW-Q — *PARTIALLY CLOSED*)

Status: Confirmed.

GIS / Map integration (07.04–22.04): Integration with the ArcGIS platform using an **iframe and URL parameters**. The solution supports equipment ID synchronization and **Azure SSO pass-through authentication**. Selected equipment IDs from portal checkboxes are passed directly to the GIS iframe.

Map visualization including **speed, heading, historical routes, sensor data, and current-day utilization** is sourced from the **Cosmos DB via API**.

Responsibilities (22.04)

- **GIS team:** Owns all map rendering, behavior, and visualization logic
- **Booking Tool team:** Provides business requirements and equipment identifiers

Tracker types and synchronization frequency (22.04):

Tracker type	Sync frequency	Notes
Teltonika (GSM)	Real-time (continuous)	Last known position shown when device is off
LoRaWAN (GPS-only)	~3 times/day	Battery-powered; low sync frequency
Additional GPS sensors	~3 times/day	Battery-powered; planned for equipment without telemetry (e.g. Kamaz)

Expected tracker data fields (IVMS, confirmed 20.04): speed , latitude , longitude , heading , engine hours , odometer (kilometres) , ignition status , battery charge .

9 Resolved Conflicts

#	Conflict	Original BRD Position	Resolution
CONFLICT-01	FO editing booking period after	FR-048: editing before confirmation	Resolved (10.04): FO can edit before AND after confirmation.

#	Conflict	Original BRD Position	Resolution
	confirmation	only	
CONFLICT-02	Booking status auto-transition by datetime	FR-070/071/075: automatic transitions	Resolved (14.04): Manual close accepted. FR-070, FR-071, FR-075 → Obsolete.
CONFLICT-03	External fleet booking workflow	FR-051..057: CC Owner → External FO	Resolved (08.04): On-demand BP = showcase only. All external booking FRs → Obsolete.

10 Open Questions

OQ ID	Description	Owner	Status
OQ-29	Dynamic filter matrix: equipment type → filter set	Team	CLOSED 22.04 — handled by dynamic characteristics admin tool; FR-NEW-39 Confirmed
OQ-36	Aggregated Request status state machine (exact mapping)	Team	CLOSED 22.04 — resolved by dev team
OQ-37	Booking Snapshot — exact field list	Team	Open
OQ-41	UI details for Unwheeled flow / manual close fields	Team	Open
OQ-NEW-A	Concrete field list for equipment cards per equipment type	TCO Fleet Owners (Asylbek)	CLOSED 22.04 — covered by dynamic characteristics admin tool; base data from TCO equipment DB
OQ-NEW-	Request / Equipment Template scope	TCO	CLOSED 22.04 — dynamic characteristics

OQ ID	Description	Owner	Status
B			pulled automatically per equipment type; no manual template selection
OQ-NEW-C	Service Work Processor role — SWP role defined: users with SWP see JDE requests; users without use Default WO. JDE trigger status (status 55 confirmation) still TBD	TCO	PARTIALLY CLOSED 22.04
OQ-NEW-E	FO delegation mechanism	TCO Cybersecurity	CLOSED 22.04 — delegation via AAD groups; no in-system feature required
OQ-NEW-F	Equipment type parameters not yet covered (Vacuum Truck and others)	TCO (Asylbek)	CLOSED 22.04 — covered by dynamic characteristics admin tool
OQ-NEW-H	Utilization booking threshold — can booking be submitted for equipment near 100% utilization?	TCO	Closed — FO decides
OQ-NEW-L	Final gradation of equipment categories and subcategories	Asylbek	Open (action from 20.04)
OQ-NEW-M	Final list of Work Centers	Asylbek	Open (action from 20.04)
OQ-NEW-N	Equipment type fields and data examples for all categories	TCO (Asylbek / colleagues)	CLOSED 22.04 — covered by dynamic characteristics admin tool
OQ-NEW-	GIS map requirements: full requirements alignment with GIS team	Duman / Team	Open — call scheduled 23.04

OQ ID	Description	Owner	Status
O	(MAPH/Atlas)		
OQ-NEW-P	JDE update frequency: DataLake jobs run once/day (~05:00). Direct JDE API preferred (target ~5 min). Pending JDE team feasibility confirmation.	Ayan / JDE team	UPDATED 22.04 — direct API proposed; TBD
OQ-NEW-Q	Utilization dashboard update frequency. Logistics: once at end of working day (previous day data) — CLOSED for Logistics . Maintenance team: requires clarification — possibly ~30–60 min for GIS map tooltip current-day utilization.	Team / Maintenance	PARTIALLY CLOSED 22.04 — Logistics OK; Maintenance TBD
OQ-NEW-R	Long-term rented equipment transferred to another department: is additional CC Owner approval needed for re-invoicing? Previously confirmed: FO manages at own discretion, no additional approval. To re-confirm at next BRD review with new shift.	Team	NEW — Open (22.04)

11 New Requirements Summary

Requirements not present in the original BRD, identified during requirements gathering sessions (April 2026):

ID	Summary	Session	Status
FR-NEW-01	All time parameters configurable via Admin Panel	03.04	Confirmed

ID	Summary	Session	Status
FR-NEW-02	FO delegation to any TCO employee with auto-revoke	08.04	Obsolete 22.04 — delegation via AAD groups outside system
FR-NEW-03	Admin configures FO timeout thresholds	03.04	Confirmed
FR-NEW-04	Admin authorizes specific users to book Assigned equipment	08.04	Confirmed — remains valid; mandatory justification added
FR-NEW-05	FO uploads multiple photos; Requestor sees them	08.04, 14.04	Confirmed
FR-NEW-06	Repair status from JDE/DataLake in search and list	08.04, 14.04	Pending
FR-NEW-07	Stationary HDE excluded from Requestor search	08.04	Confirmed
FR-NEW-08	Assigned equipment: visible to all, bookable by Admin-authorized users with mandatory justification; FO can decline	08.04	Confirmed
FR-NEW-09	Shared with Conditions has visual color marker	03.04	Confirmed
FR-NEW-10	On-demand only; no weekly schedules	03.04	Confirmed

ID	Summary	Session	Status
FR-NEW-11	Priority P1–P4 selected by Requestor; WO-level attribute	03.04, 17.04	Confirmed
FR-NEW-12	WO mandatory for Maint/Railroad/Ops; optional for Logistics; Location for SCM	08.04, 16.04	Confirmed
FR-NEW-13	"Work Description" mandatory (~50 chars)	08.04	Confirmed
FR-NEW-14	"Comments" optional	08.04	Confirmed
FR-NEW-15	Single form → auto-split into atomic bookings	09.04	Confirmed
FR-NEW-16	Partial confirmation per RequestItem	03.04	Confirmed
FR-NEW-17	Aggregated Request status auto-calculated	09.04	Confirmed 22.04 (OQ-36 Closed)
FR-NEW-18	Booking start = mobilization start	03.04	Confirmed
FR-NEW-19	No mandatory buffer between bookings	09.04	Confirmed
FR-NEW-20	Unwheeled equipment warning in UI	07.04	Preliminary

ID	Summary	Session	Status
FR-NEW-21	FO timeout: 24h reminder / 48h escalation	03.04, 06.04	Confirmed
FR-NEW-22	FO must state reason for decline if equipment available	06.04	Confirmed
FR-NEW-23	Reminder only to FO of specific equipment	07.04	Confirmed
FR-NEW-24	FO "Mobilization started" button	09.04	Confirmed
FR-NEW-25	BP showcase: view-only catalog	08.04	Confirmed
FR-NEW-26	BP prices not displayed	08.04	Confirmed
FR-NEW-27	BP self-populates catalog cards	08.04	Confirmed
FR-NEW-28	"Go to BP catalog" button after FO timeout	08.04	Confirmed
FR-NEW-29	Three-step Unwheeled flow with Transportation role	09.04	Confirmed
FR-NEW-30	FO proposes nearest date if transport unavailable	03.04	Pending

ID	Summary	Session	Status
FR-NEW-31	Early close button for Requestor/FO	07.04	Pending
FR-NEW-32	Manual close only	09.04	Confirmed
FR-NEW-33	Actual start/end time at close for utilization	09.04	Confirmed
FR-NEW-34	FO reminder 24h / escalation 48h	03.04, 06.04	Confirmed
FR-NEW-35	Notification to Transportation Responsible	09.04	Confirmed
FR-NEW-36	FO notified of Transportation Responsible decision	09.04	Confirmed
FR-NEW-37	Audit trail on demand	07.04	Confirmed
FR-NEW-38	Search: TCO number + model mandatory; гоcнoмep conditional	08.04, 13.04	Confirmed
FR-NEW-39	Dynamic search filters by equipment type (via dynamic characteristics admin tool)	08.04 / Confirmed 22.04	Confirmed
FR-NEW-40	Booking calendar in equipment card (FO view)	13.04	Confirmed

ID	Summary	Session	Status
FR-NEW-41	Multiple trackers per unit; add from UI	13.04, 14.04	Confirmed
FR-NEW-42	Shared team email per fleet (Admin-managed)	13.04	Confirmed
FR-NEW-43	Notifications to fleet shared team email	13.04	Confirmed
FR-NEW-44	Request → Completed when last active booking closed	14.04	Confirmed
FR-NEW-45	"Completed Requests" page	14.04	Confirmed
FR-NEW-46	Utilization reports by day / department / equipment	14.04	Pending
FR-NEW-47	Work center reports	14.04, 16.04	Pending
FR-NEW-48	"Выбрать WO из JDE" block; auto-generate draft	16.04	Confirmed
FR-NEW-49	Freeze fields in equipment card	16.04	Confirmed
FR-NEW-50	Admin manages all reference/handbook values	16.04	Confirmed

ID	Summary	Session	Status
FR-NEW-51	Default WO / Default WC for non-JDE users (Logistics and others)	17.04	Confirmed
FR-NEW-52	"История заявок" page for Requestor (completed requests history)	17.04	Confirmed
FR-NEW-53	Utilization dashboard — 3 tabs: Моточасы / Километраж / Комбайн	17.04	Confirmed
FR-NEW-54	Utilization cells — color gradient: 0% red → 50% yellow → 100% green	17.04	Confirmed
FR-NEW-55	Repair days — wrench icon; JDE fields (commitment/planned/completed date); excluded from average	17.04	Confirmed (manual)
FR-NEW-56	Utilization dashboard — Target value column (FO-defined max per equipment)	17.04	Confirmed
FR-NEW-57	Utilization dashboard — Average column; sortable by average	17.04	Confirmed
FR-NEW-58	Utilization dashboard — Day/Month view toggle	17.04	Confirmed
FR-NEW-59	Utilization dashboard — Ownership filter (TCO Owned / Long-term rented)	17.04	Confirmed
FR-NEW-60	Utilization from trackers only; independent of bookings	17.04	Confirmed

ID	Summary	Session	Status
FR-NEW-61	GIS map — iframe; checkboxes in equipment list; TSO search; expand button	17.04	Confirmed (GIS team)
FR-NEW-62	GIS map tooltip — sensor data from last sync + current-day utilization (Maintenance requirement); refresh interval TBD	17.04 / Updated 22.04	Pending (GIS team + interval)
FR-NEW-63	Utilization dashboard — booking count for selected period next to average utilization	20.04	Confirmed
FR-NEW-64	FO approval window — equipment loading summary by dates (list of nearby active bookings)	20.04	Confirmed
FR-NEW-65	GIS map — speed and heading for GSM-tracked equipment; heading not shown for LoRaWAN	20.04	Confirmed
FR-NEW-66	GIS map — fuel level not displayed	20.04	Confirmed
FR-NEW-67	GIS map — historical route for 7 days	20.04	Confirmed (GIS team)
FR-NEW-68	Admin Panel tool for configuring dynamic custom characteristics per equipment type; system shows only type-specific characteristics in request form and search filters	22.04	Confirmed